**DISCOVER · wide-field surveys find $\mathcal{O}(1000)$ lensed AGN**

**CHARACTERIZE**

## Rubin LSST

PSF FWHM 0.71–0.77″

**g r i** · 10-yr

## Euclid

PSF FWHM 0.16–0.5″

**VIS Y J H** · wide

## Roman

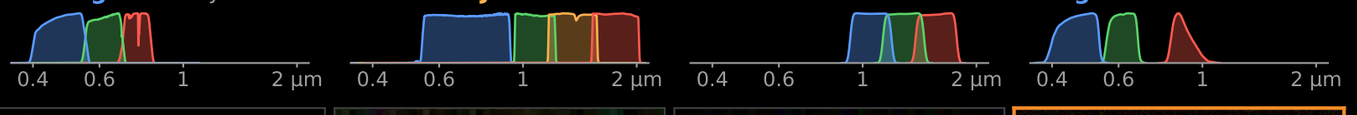
PSF FWHM 0.087–0.13″

**F106 F129 F158** · HLWAS

## Lazuli WCC

PSF FWHM 33–62 mas

**LZg LZr LZz** · 4×300 s




**Quad** $z_d$=0.89 $z_s$=2.74

$\theta_E$ = 0.84 arcsec

1 arcsec

IFS field

D
G
C
A
B

**Quad** $z_d$=1.16 $z_s$=2.64

$\theta_E$ = 0.23 arcsec

1 arcsec

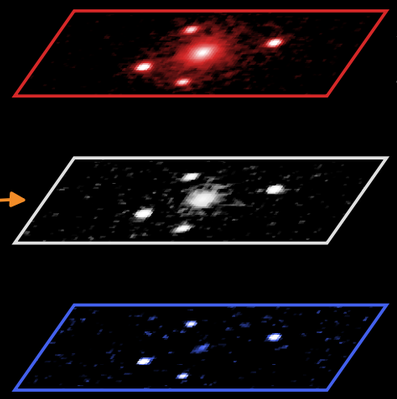
**Double** $z_d$=0.57 $z_s$=1.22

$\theta_E$ = 0.74 arcsec

1 arcsec

## LAZULI IFS · one pointing, every image

0.4–1.7 μm · R ≈ 100–500 · 40 mas spaxels · 3 h

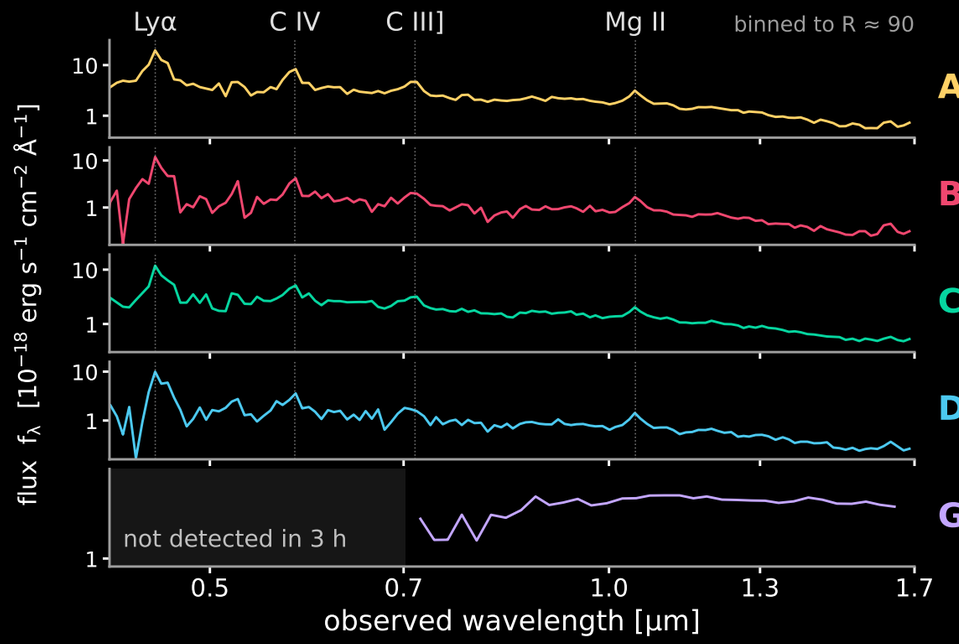



**1.20–1.50 μm**
lens galaxy: red starlight

**0.85–1.00 μm**
lens galaxy appears

**0.60–0.70 μm**
quasar images

**Same quasar lines in every image → lensing confirmed**




Simulated with SLSim; quasar hosts from HST COSMOS. Lensed-AGN counts: Oguri & Marshall 2010.
Lazuli: Roy et al. 2026 (arXiv:2601.02556). Modelled, not official: diffraction-limited 3.06 m Airy PSF + 4 mas jitter, SDSS-shaped filters, throughput, noise, exposure times.

Credits: Paras Sharma (Stony Brook University)